Modelling Software-based Systems

Correctness by Contruction of sequential recursive algorithms using Event-B Refinement

Master Informatique

Dominique Méry
Telecom Nancy,Université de Lorraine

4 novembre 2026

dominique.mery@loria.fr

Summary

1. CM5
2. The Call-as-Event Paradigm
3. Two examples
   - Computing the square of a number
   - Searching key in an array

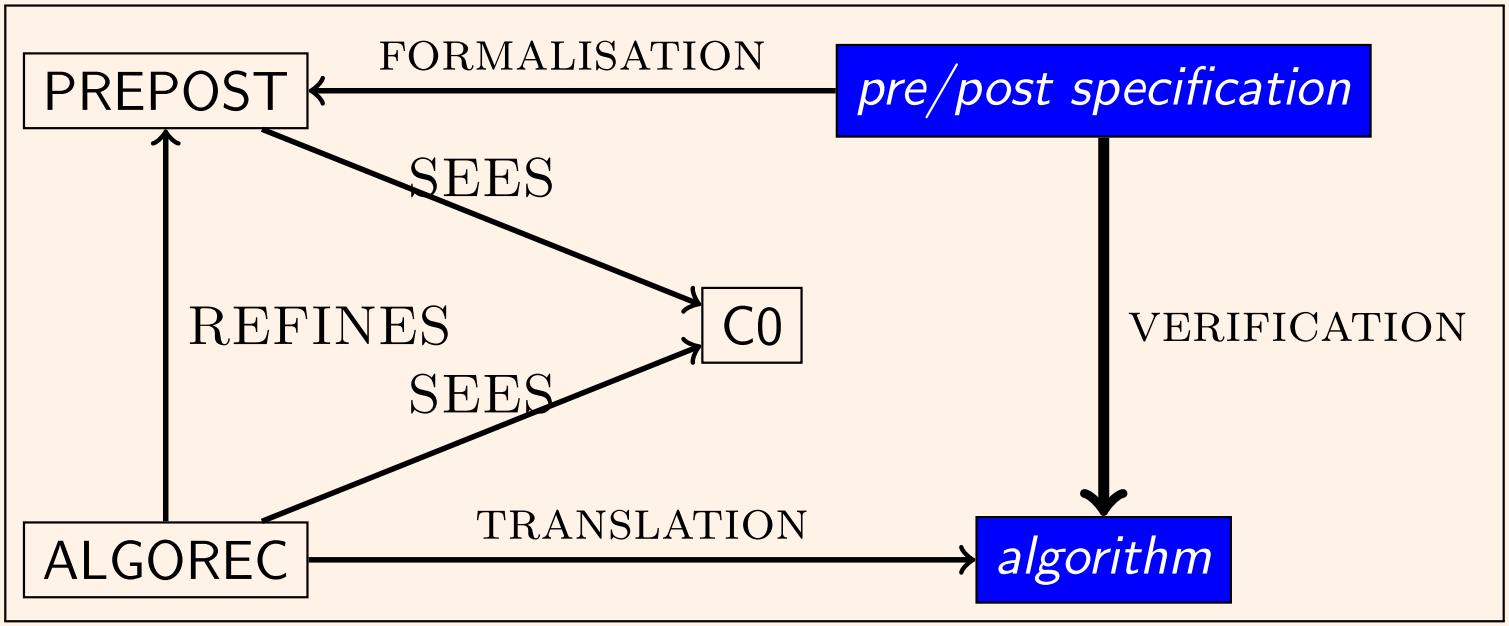
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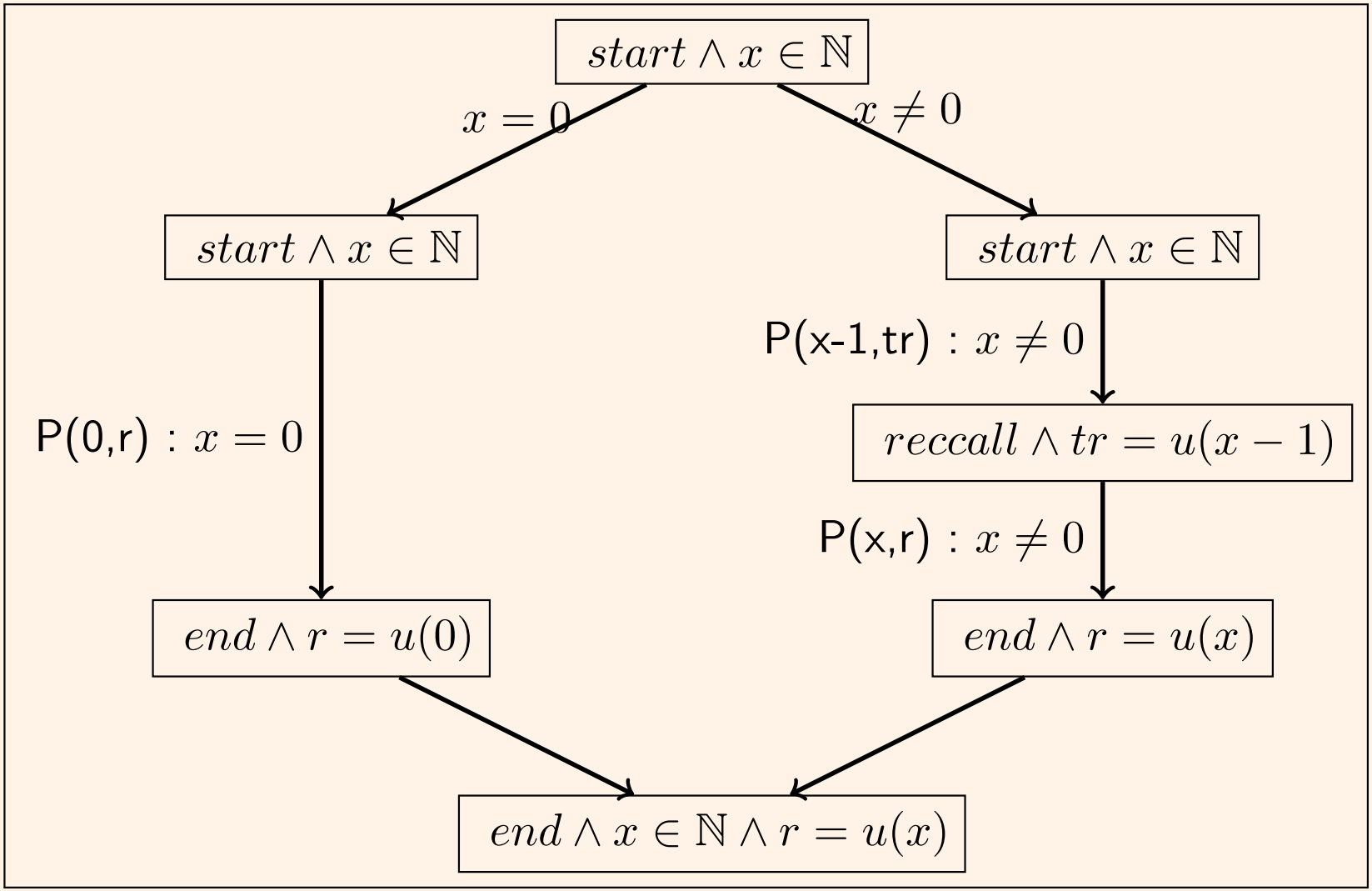
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The recursive pattern



The refinement diagram



Ideas for producing algorithms

- **EVENT** rec%PROC(h(x),y)%P(y) is simply simulating the recursive call of the same function.
- The invariant is defined in a simpler way by analysing the inductive structure and a control variable is introduced for structuring the inductive computation.

EVENT
e
WHERE
$\ell = \ell_1$
$g_{\ell_1,\ell_2}(x)$
THEN
$\ell := \ell_2$
$x := f_{\ell_1,\ell_2}(x)$
END

EVENT
rec%PROC(h(x),y)%P(y)
ANY y
WHERE
$\ell = \ell_1$
$g_{\ell_1,\ell_2}(x, y)$
THEN
$\ell := \ell_2$
$x := f_{\ell_1,\ell_2}(x, y)$
END

EVENT
call%APROC(h(x),y)%P(y)
ANY y
WHERE
$\ell = \ell_1$
$g_{\ell_1,\ell_2}(x, y)$
THEN
$\ell := \ell_2$
$x := f_{\ell_1,\ell_2}(x, y)$
END

The computation of the function x^2

VARIABLES

r, c, tr

INVARIANTS

$inv1 : r \in \mathbb{N}$

$inv2 : c = end \Rightarrow r = n * n$

$inv3 : c = callrec \Rightarrow n \neq 0$

$inv4 : c = callrec \Rightarrow tr = (n-1) * (n-1)$

$inv5 : c \in C$

$inv6 : tr \in \mathbb{N}$

$inv7 : c = end \Rightarrow r = n * n$

$inv8 : c = end \wedge n \neq 0$
$\Rightarrow tr = (n-1) * (n-1) \wedge r = tr + 2 * (n-1) + 1$

$inv9 : c = callrec \Rightarrow n * n = tr + 2 * (n-1) + 1$

The computation of the function x^2

EVENT INITIALISATION
BEGIN
$act1 : r := 0$
$act2 : c := start$
$act3 : tr :\in \mathbb{N}$
END
EVENT square0
REFINES square(n ;r)
WHEN
$grd1 : c = start$
$grd2 : n = 0$
THEN
$act1 : c := end$
$act2 : r := 0$
END

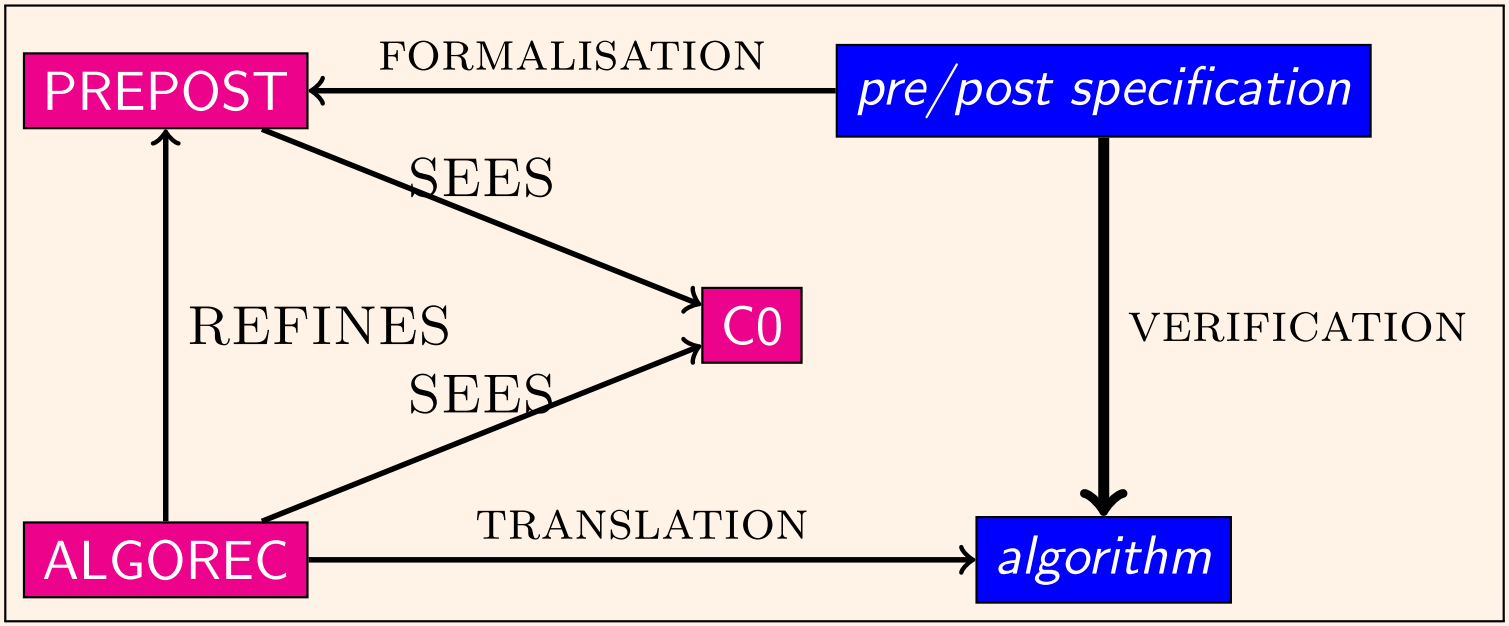
EVENT squaren
REFINES square(n ;r)
WHEN
$grd1 : c = callrec$
THEN
$act1 : r := tr + 2 * (n - 1) + 1$
$act2 : c := end$
END
EVENT rec%square(n-1 ;tr)
WHEN
$grd1 : c = start$
$grd2 : n \neq 0$
THEN
$act1 : c := callrec$
$act2 : tr := (n - 1) * (n - 1)$
END

Summary for proof obligations

Summary for proof obligations

Name	Total	Automatic	Interactive
cae-square	34	32	2
square0	3	2	1
specquare	2	2	0
square	29	28	1

The recursive pattern



Comments

- Proofs are easier and simpler
- Invariant is simple to find
- Translation is automatic
- Students are not happy with it and tools are not always set for this verification.

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